Nikul Patel

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EDUCATION

Clarkson University — Potsdam, NY
Double Major: Aerospace and Mechanical Engineering | Minor: Robotics | Honors Program — *Expected Graduation: 2028*
Clubs and Organizations: Society of Asian Scientists and Engineers | Underwater Robotics — GPA: 3.96

PROFESSIONAL EXPERIENCE

*Link to Projects embedded in Digital Version.

Design Build Captain — Aug 2025 – Present
Engineering Design Team — *Potsdam, NY*

- Designed and 3D-printed custom mounts and fixtures for cameras, IMUs, pressure sensors, and control boards using SolidWorks and OnShape to ensure reliability in underwater environments.
- Worked closely with electrical and controls teams to integrate wiring, connectors, and microcontrollers into mechanical layouts for clean, serviceable robot builds.

Undergraduate Research Assistant — Jan 2025 – May 2025
Photoacoustic Research Laboratory — *Potsdam, NY*

- Designed and 3D-printed mechanical fixtures to mount and align ultrasonic sensors for automated data collection, improving measurement stability and repeatability by 20%.
- Worked with controls engineering students to integrate hardware with PLC systems and ran iterative system tests to improve reliability, throughput, and scalability.

Research Intern* — Mar 2024 – July 2024
Spartificial — *Remote*

- Analyzed and organized a dataset of 22,000+ samples across 22 classes and 4 timeframes; independently designed a Late Fusion multi-view CNN while leading a 2-member team on Early &Intermediate Fusion models.
- Achieved 95%+ accuracy for each model and 20% faster training , while optimization experiments improved predictive accuracy by 10% and processing efficiency by 15%.

Aerospace Structures Design Intern* — Oct 2023 – Dec 2023
Feynman Aerospace — *Mumbai, IND*

- Designed and assembled a 1U CubeSat structural model in SolidWorks, incorporating deployable antennas, solar panels, and frame assemblies; conducted comparative material evaluation (Al 7075-T6, CFRP, Ti alloys) to balance strength-to-weight and manufacturability.
- Performed finite element static and thermal load analyses in Ansys under launch and orbital worst-case scenarios, demonstrating a possibility of 25% increase in structural reliability and a 15% reduction in material usage.

LEADERSHIP AND ACADEMIC ENGAGEMENT

Teaching Assistant | MATLAB (HP102/103) | Clarkson University — *Aug 2025 – Present*

- Co-Managing a fully TA-led honors course | Coordinating lectures, curriculum, assignments, grading, and office hours.

Tutor | Introductory Engineering Courses | Clarkson University — *Aug 2025 – Present*

- Supported fellow students with homework, and clarified difficult topics, throughout drop-in and small group sessions.

Class Senator | Student Association | Clarkson University — *Sept 2025 – Present*

- Advocating for student interests through active participation in policy discussions and voting on key campus issues.

Class Representative | SASE (Society of Asian Scientists and Engineers) | Clarkson Chapter — *Sept 2024 – Present*

- Representing my class, organized cultural and professional events, led fundraisers, promoting Asian heritage.

Admissions Ambassador | Undergraduate Admissions | Clarkson University — *Sept 2024 – Present*

- Giving campus tours, performing outreach, assisting with administrative tasks, and answering front desk calls.

SKILLS

Programming Languages: MATLAB, Python, Arduino (embedded systems), ROS (basic), C/C++ (microcontrollers)
Software and Cloud Platforms: SolidWorks, Ansys, Visual Studio Code, Microsoft Office Suite, Kaggle, Google Collab
Hardware & Robotics: CAD design, 3D printing, mechanical prototyping, sensor integration, and hardware debugging

AWARDS

Presidential Scholar: Academic Excellence | *Clarkson University* — *Fall 2024, 2025 & Spring 2025*
SASE Rising Leader: Recognition for strong leadership skills and proactive initiative | *Clarkson University* — *April 2025*
Master BSMS: Nominated Best student of outgoing batch in all aspects. | *B.S. Memorial School* — *Feb 2023*